School of Computer Science and Artificial Intelligence

Lab Assignment - 19.3

Course Name : AI Assistant Coding
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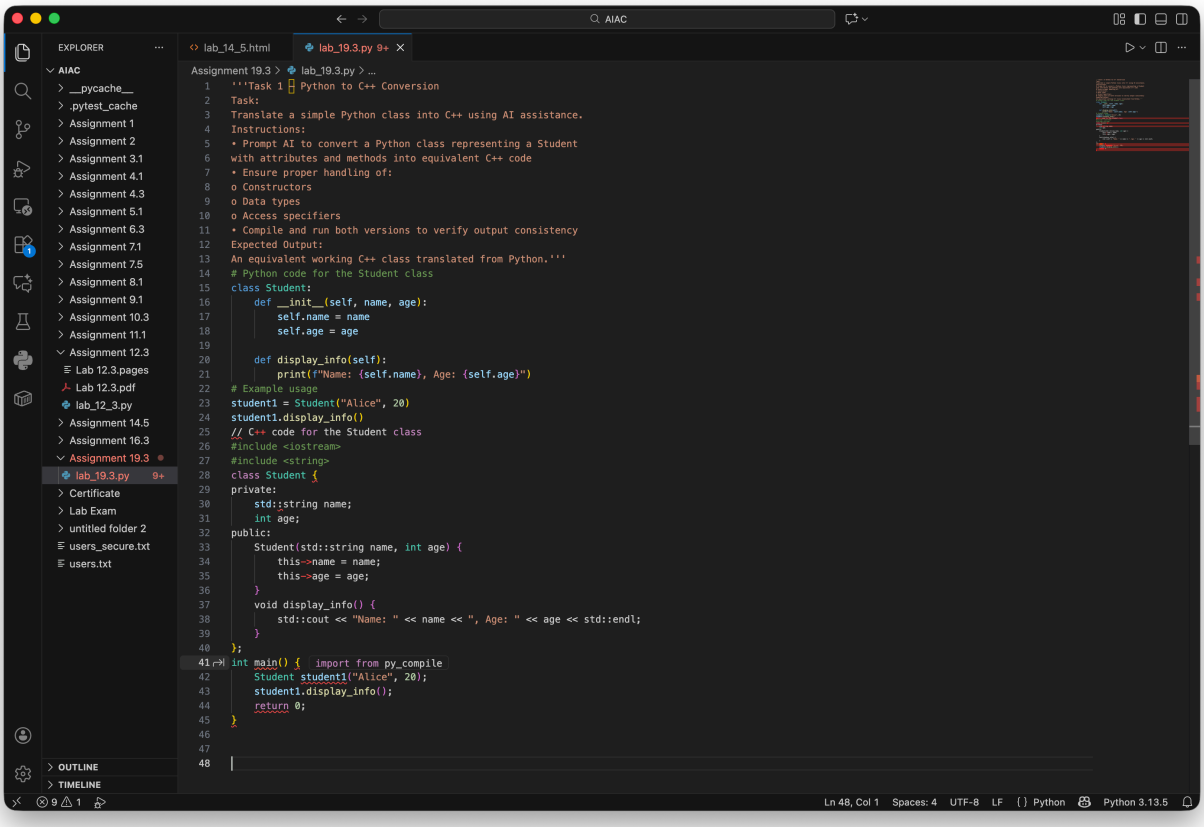
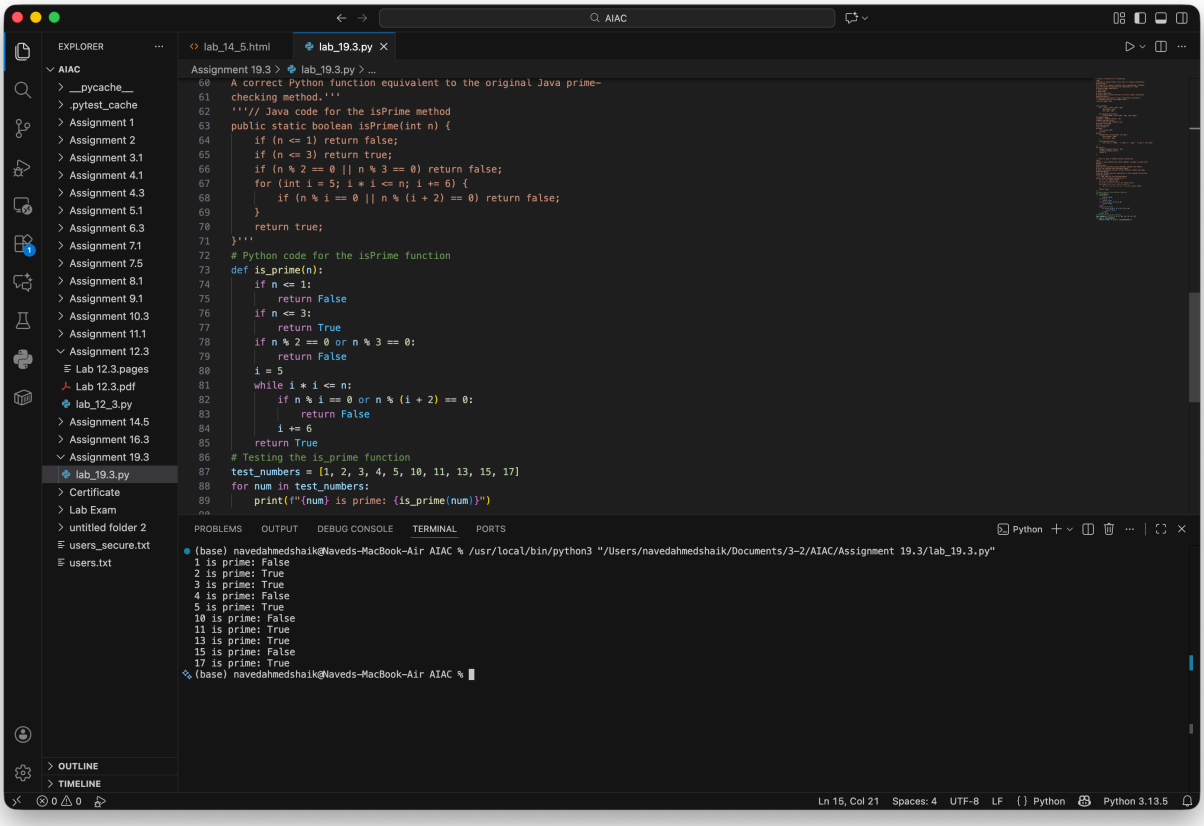
Batch No : 47-B

Task 1 – Python to C++ Conversion
Task:
Translate a simple Python class into C++ using AI assistance.
Instructions:
• Prompt AI to convert a Python class representing a Student with attributes and methods into equivalent C++ code
• Ensure proper handling of:
o Constructors
o Data types
o Access specifiers
• Compile and run both versions to verify output consistency
Expected Output:
An equivalent working C++ class translated from Python.



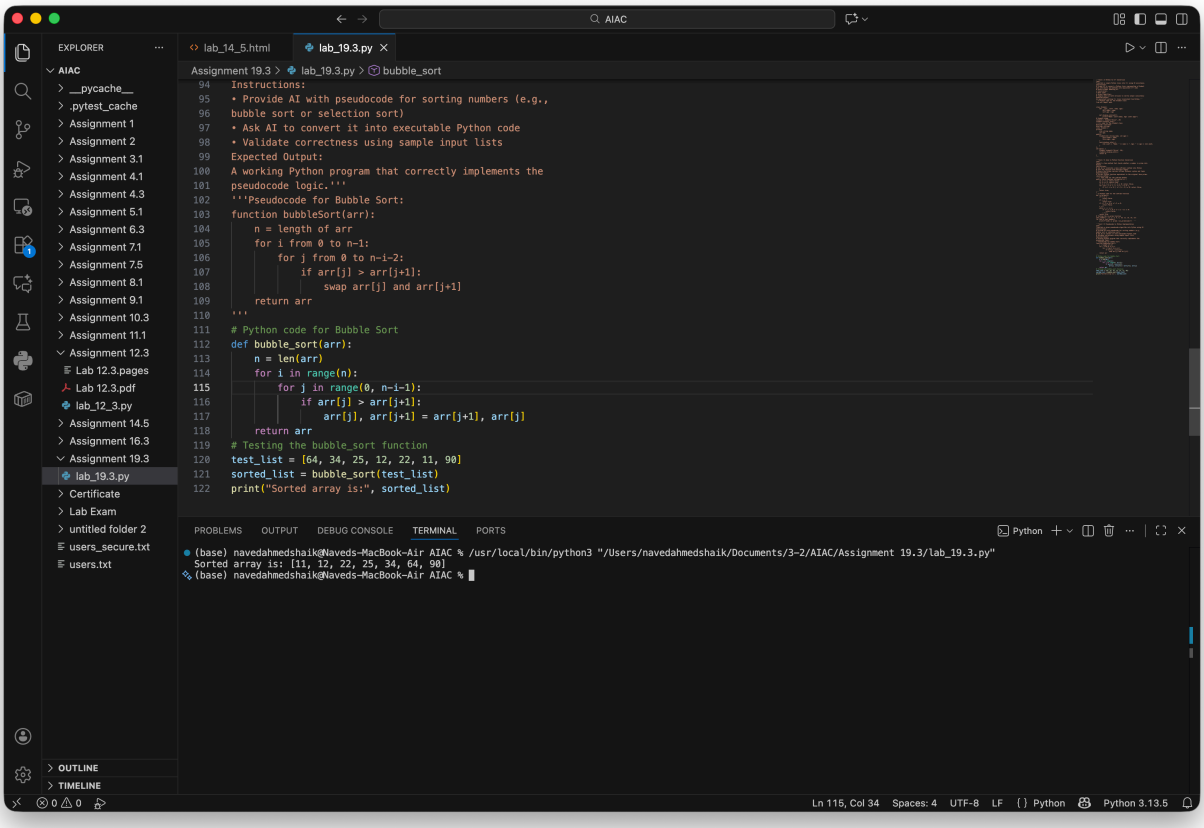
Task 2 – Java to Python Function Conversion
Task:
Convert a Java method that checks whether a number is prime into Python.
Instructions:
- **Ask AI to translate a Java isPrime() method into Python**
- **Test the function with multiple inputs**
- **Ensure the Python version follows Pythonic syntax and logic**

Expected Output:
A correct Python function equivalent to the original Java prime-checking method.

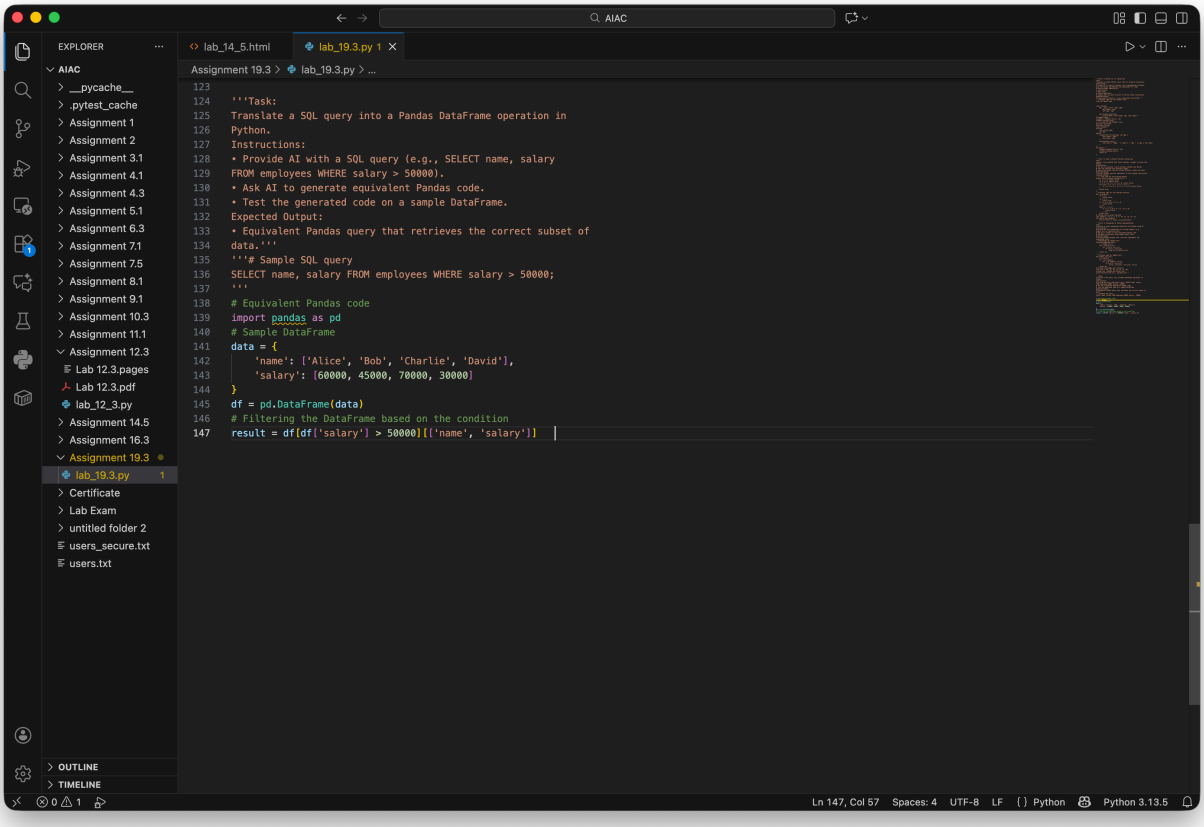


Task 3 – Pseudocode to Python Implementation
Task:
Translate a given pseudocode algorithm into Python using AI.
Instructions:
- **Provide AI with pseudocode for sorting numbers (e.g., bubble sort or selection sort)**
- **Ask AI to convert it into executable Python code**
- **Validate correctness using sample input lists**

Expected Output:
A working Python program that correctly implements the pseudocode logic.


AIAC
EXPLORER
lab_14_5.html
lab_19.3.py
Assignment 19.3 > lab_19.3.py > bubble_sort
__pycache__
.pytest_cache
Assignment 1
Assignment 2
Assignment 3.1
Assignment 4.1
Assignment 4.3
Assignment 5.1
Assignment 6.3
Assignment 7.1
Assignment 7.5
Assignment 8.1
Assignment 9.1
Assignment 10.3
Assignment 11.1
Assignment 12.3
Lab 12.3.pages
Lab 12.3.pdf
lab_12_3.py
Assignment 14.5
Assignment 16.3
Assignment 19.3
lab_19.3.py
Certificate
Lab Exam
untitled folder 2
users_secure.txt
users.txt
OUTLINE
TIMELINE
Instructions:
• Provide AI with pseudocode for sorting numbers (e.g.,
bubble sort or selection sort)
• Ask AI to convert it into executable Python code
• Validate correctness using sample input lists
Expected Output:
A working Python program that correctly implements the
pseudocode logic.'''
'''Pseudocode for Bubble Sort:
function bubbleSort(arr):
n = length of arr
for i from 0 to n-1:
for j from 0 to n-i-2:
if arr[j] > arr[j+1]:
swap arr[j] and arr[j+1]
return arr
'''
# Python code for Bubble Sort
def bubble_sort(arr):
n = len(arr)
for i in range(n):
for j in range(0, n-i-1):
if arr[j] > arr[j+1]:
arr[j], arr[j+1] = arr[j+1], arr[j]
return arr
# Testing the bubble_sort function
test_list = [64, 34, 25, 12, 22, 11, 90]
sorted_list = bubble_sort(test_list)
print("Sorted array is:", sorted_list)
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
Python
(base) navedahmedshaik@Naveds-MacBook-Air AIAC % /usr/local/bin/python3 "/Users/navedahmedshaik/Documents/3-2/AIAC/Assignment 19.3/lab_19.3.py"
Sorted array is: [11, 12, 22, 25, 34, 64, 90]
(base) navedahmedshaik@Naveds-MacBook-Air AIAC %
Ln 115, Col 34
Spaces: 4
UTF-8
LF
Python
Python 3.13.5

Task 4 – SQL to Pandas Query
Task:
Translate a SQL query into a Pandas DataFrame operation in Python.
Instructions:
• Provide AI with a SQL query (e.g., SELECT name, salary FROM employees WHERE salary > 50000).
• Ask AI to generate equivalent Pandas code.
• Test the generated code on a sample DataFrame.
Expected Output:
• Equivalent Pandas query that retrieves the correct subset of data.



Task 5 – Real-Time Application: Algorithm Translation Across Languages

Scenario:

A company maintains algorithms written in different programming languages.

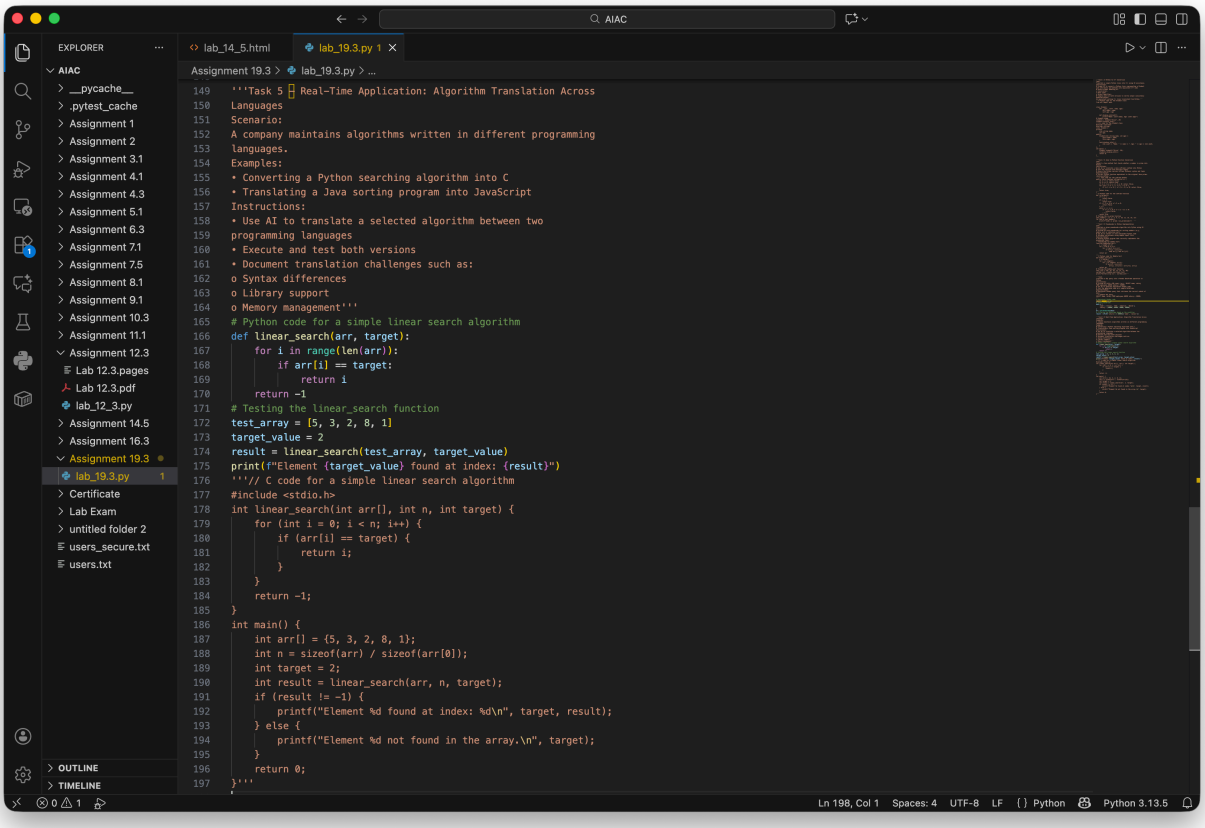
Examples:

- **Converting a Python searching algorithm into C**
- **Translating a Java sorting program into JavaScript**

Instructions:

- **Use AI to translate a selected algorithm between two programming languages**
- **Execute and test both versions**
- **Document translation challenges such as:**

o Syntax differences

o Library support

o Memory management